HTML for Beginners

A Complete Step-by-Step Tutorial

From Absolute Zero to Building Your First Websites

What You Will Learn

- Complete HTML fundamentals with clear explanations
- Every important tag with live code examples
- How to structure pages using div, span & semantic tags
- Practice sets after every chapter
- Important interview-style questions
- Real-world tips and best practices

Perfect for Absolute Beginners

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Chapter 0 – Introduction

Welcome to the World of Web Development!

Congratulations on taking the first step toward becoming a web developer! This book is written especially for absolute beginners. You do not need any previous programming experience. We will start from zero and build solid foundations step by step.

By the end of this tutorial you will be able to create complete HTML pages, understand how websites are structured, and write clean, readable code that browsers can understand.

Why HTML, CSS & JavaScript?

A modern website is built using three core technologies. Think of them as three layers that work together:

1. HTML (HyperText Markup Language) – The Structure

HTML is the skeleton of a webpage. It defines the content and structure: headings, paragraphs, images, links, buttons, forms, etc. Without HTML there is nothing for the browser to display. HTML uses tags to mark up content so the browser knows what each piece means.

2. CSS (Cascading Style Sheets) – The Design

CSS is responsible for how the page looks: colors, fonts, spacing, layouts, animations, and responsiveness. HTML provides the content; CSS makes it beautiful. You can change the entire look of a website without touching the HTML.

3. JavaScript – The Behavior

JavaScript makes websites interactive. It handles clicks, form validation, dynamic content updates, animations, and communication with servers. HTML is structure, CSS is style, JavaScript is the brain that makes things happen.

> **TIP:** In this book we focus only on HTML. Once you master HTML, learning CSS and JavaScript becomes much easier because you already understand the structure of a page.

A Beautiful Analogy

Imagine you are building a house:

- **HTML** is the foundation, walls, rooms, doors and windows — the structure.
- **CSS** is the paint, wallpaper, furniture arrangement and lighting — the design.
- **JavaScript** is the electricity, plumbing, and smart home features — the behavior.

You cannot paint a house that has no walls. That is why we start with HTML. A solid HTML foundation makes everything else easier and more reliable.

Installing VS Code

VS Code (Visual Studio Code) is the most popular free code editor used by professional developers worldwide. It is lightweight, powerful, and has excellent support for HTML.

Step-by-step installation:

1. Open your browser and go to **https://code.visualstudio.com**
2. Click the big download button for your operating system (Windows / Mac / Linux).
3. Run the installer and follow the simple on-screen instructions.
4. After installation, open VS Code.
5. Create a new folder on your computer called **html-practice**.
6. In VS Code go to File → Open Folder and select that folder.
7. Create a new file called **index.html** (File → New File, then save as index.html).

TIP: Recommended VS Code extensions for HTML: Live Server (by Ritwick Dey) – lets you see changes instantly in the browser. Prettier – formats your code automatically. HTML CSS Support – gives autocomplete.

Once you have VS Code ready and a blank index.html file open, you are ready to write your first HTML code!

Chapter 1 – Creating Our First Website

A Basic HTML Page

Every HTML document follows a standard structure. Let us write the smallest possible complete webpage.

Example 1.1 – Your First HTML Page

```
<!DOCTYPE html>
<html>
<head>
<title>My First Website</title>
</head>
<body>
<h1>Hello World!</h1>
<p>This is my first webpage. I am learning HTML.</p>
</body>
</html>
```

Save this file as **index.html** and open it in any browser (Chrome, Firefox, Edge). You will see a heading and a paragraph.

Understanding Every Line

<!DOCTYPE html> — This declaration tells the browser that this document is written in HTML5 (the modern version of HTML). Always put it on the very first line.

<html> ... </html> — The root element. Everything else goes inside these tags.

<head> ... </head> — Contains meta information about the page that is not directly visible. The title, character encoding, CSS links, and scripts usually go here.

<title> ... </title> — Sets the text that appears in the browser tab. Also used by search engines.

<body> ... </body> — Contains everything that is visible on the page: text, images, links, buttons, etc.

Important Notes

HTML is made of **elements**. Most elements have an opening tag and a closing tag.

Opening tag: `<p>` Closing tag: `</p>`

Whatever you write between the opening and closing tags is the content of that element.

> **IMPORTANT:** Some tags are self-closing (also called void elements). They do not need a closing tag. Examples: <br>, <hr>, <img>, <input>, <meta>.

Browsers are very forgiving. Even if you make small mistakes, the page often still works. However, writing correct HTML is important for accessibility, SEO, and maintainability.

Comments in HTML

Comments are notes for humans. The browser completely ignores them. Use comments to explain your code or temporarily disable parts of it.

Example 1.2 – HTML Comments

```
<!-- This is a single-line comment -->

<!--
This is a multi-line comment.
You can write as many lines as you want.
-->

<p>This paragraph is visible.</p>
<!-- <p>This paragraph is hidden because it is inside a comment.</p> -->
```

TIP: Good developers leave clear comments in complex sections. But do not over-comment obvious code. Comment the 'why', not the 'what'.

Case Sensitivity

HTML tags are **not** case-sensitive. <P>, <p> and <P> are treated the same by browsers. However, the modern standard and best practice is to write all tags in **lowercase**.

Example 1.3 – Preferred Style

```
<!-- Good practice (lowercase) -->
<p>Hello</p>
<img src="photo.jpg" alt="A photo">

<!-- Avoid this (mixed or uppercase) -->
<P>Hello</P>
<IMG SRC="photo.jpg" ALT="A photo">
```

Attribute values can be written with double quotes or single quotes. Double quotes are more common and recommended.

Chapter 1 – Practice Set

Practice 1: Create a file named **about-me.html**. Write a complete HTML page with the title “About Me”. Inside the body put a heading with your name and two paragraphs describing yourself.

Practice 2: Add at least three meaningful comments in your about-me.html file explaining different sections.

Practice 3: Create another file called **favorite-foods.html**. Give it a proper title and write a short introduction paragraph about your favorite foods.

Practice 4: Intentionally write some tags in uppercase and some in lowercase. Open the page in the browser and observe that it still works. Then convert everything to lowercase (best practice).

Practice 5: Open your HTML file in three different browsers (Chrome, Firefox, Edge) and confirm it looks the same.

Important Questions – Chapter 1

Q1. What does HTML stand for?

Q2. What is the purpose of the <!DOCTYPE html> declaration?

Q3. What is the difference between the <head> and <body> sections?

Q4. Why should we write HTML tags in lowercase even though HTML is case-insensitive?

Q5. How do you write a multi-line comment in HTML?

Q6. Name three self-closing (void) tags.

Q7. What appears in the browser tab? Which tag controls it?

Q8. Is it mandatory to close every HTML tag? Explain with examples.

Chapter 2 – Basic HTML Tags

HTML Elements

An HTML element usually consists of a start tag, content, and an end tag.

```
<tagname>Content goes here</tagname>
```

Example: <h1>Welcome</h1> — Here <h1> is the start tag, “Welcome” is the content, and </h1> is the end tag. The whole thing is an h1 element.

HTML Attributes

Attributes provide extra information about an element. They are always written inside the opening tag and usually come in name="value" pairs.

```
<tagname attribute="value">Content</tagname>
```

Common examples you will see often:

- **src** – source of an image or script
- **href** – hyperlink reference (URL)
- **alt** – alternative text for images
- **id** – unique identifier for an element
- **class** – class name(s) for styling or JavaScript
- **title** – extra information shown as a tooltip

The Heading Tags (h1 to h6)

Headings define the hierarchy of your content. There are six levels. **h1** is the most important (usually the main title of the page). **h6** is the least important.

Example 2.1 – All Heading Levels

```
<h1>This is Heading 1 – Main Title</h1>
<h2>This is Heading 2 – Section Title</h2>
<h3>This is Heading 3 – Sub-section</h3>
<h4>This is Heading 4</h4>
<h5>This is Heading 5</h5>
<h6>This is Heading 6 – Smallest</h6>
```

TIP: Use only one h1 per page for best SEO practice. Use headings in order (do not jump from h1 directly to h4). Headings help both users and search engines understand the structure of your content.

The Paragraph Tag

The <p> tag is used for paragraphs of text. Browsers automatically add some space before and after paragraphs.

Example 2.2 – Paragraphs

```
<p>This is the first paragraph. It can contain multiple sentences.
Browsers will wrap the text according to the width of the screen.</p>

<p>This is the second paragraph. Notice the space between paragraphs.</p>
```

The Anchor Tag (Links)

The <a> tag creates hyperlinks. The most important attribute is **href** (hypertext reference).

Example 2.3 – Different Types of Links

```
<!-- Link to another website -->
<a href="https://www.google.com">Visit Google</a>

<!-- Link to another page on the same website -->
<a href="about.html">About Us</a>

<!-- Link to an email address -->
<a href="mailto:hello@example.com">Send Email</a>

<!-- Link to a phone number -->
<a href="tel:+919876543210">Call Us</a>

<!-- Link to a specific section on the same page -->
<a href="#section2">Go to Section 2</a>
```

The Image Tag

The <img> tag displays images. It is a self-closing tag. Two attributes are essential: **src** (path to the image) and **alt** (alternative text).

Example 2.4 – Adding Images

```
<!-- Image from the same folder -->
<img src="photo.jpg" alt="A beautiful sunset">

<!-- Image from a sub-folder -->
<img src="images/logo.png" alt="Company Logo">

<!-- Image from the internet (absolute URL) -->
<img src="https://example.com/picture.jpg" alt="Online image">

<!-- Setting width and height -->
<img src="photo.jpg" alt="Sunset" width="400" height="300">
```

IMPORTANT: Always provide a meaningful alt attribute. It is crucial for accessibility (screen readers) and SEO. If the image fails to load, the alt text is shown instead.

Bold, Italic and Underline Tags

These tags change the appearance of text. Modern HTML prefers semantic tags over purely presentational ones.

Example 2.5 – Text Formatting

```
<!-- Bold -->
<strong>This text is important (bold)</strong>
<b>This is also bold (older style)</b>

<!-- Italic -->
<em>This text is emphasized (italic)</em>
<i>This is also italic (older style)</i>

<!-- Underline -->
<u>This text is underlined</u>

<!-- Combined -->
<p>This is <strong>bold</strong>, this is <em>italic</em>,
and this is <u>underlined</u>.</p>
```

TIP: Prefer over <b> and over <i> because they carry meaning (importance / emphasis) rather than just visual style.

The br Tag (Line Break)

The <br> tag inserts a single line break. It is self-closing. Use it when you want to break a line without starting a new paragraph.

Example 2.6 – Line Breaks

```
<p>John Doe<br>
123 Main Street<br>
New Delhi, India<br>
PIN 110001</p>
```

The hr Tag (Horizontal Rule)

The <hr> tag creates a horizontal line (thematic break). It is useful to separate sections of content.

Example 2.7 – Horizontal Rule

```
<h2>Introduction</h2>
<p>Some introductory text...</p>
<hr>
<h2>Main Content</h2>
<p>The main content starts here...</p>
```

Big and Small Tags

The <big> tag makes text larger and <small> makes text smaller. Note that <big> is obsolete in HTML5; use CSS instead. <small> is still valid and often used for fine print or side comments.

Example 2.8 – Big & Small

```
<p>Normal text. <big>This looks bigger.</big>

<small>This looks smaller – often used for copyright text.</small></p>
```

Subscript & Superscript

Subscript (<sub>) appears slightly below the baseline. Superscript (<sup>) appears slightly above. Useful for chemical formulas, mathematical expressions, and footnotes.

Example 2.9 – Sub and Sup

```
<p>Water formula: H<sub>2</sub>O</p>

<p>Einstein's equation: E = mc<sup>2</sup></p>

<p>The 1<sup>st</sup> of January, 2025</p>

<p>Footnote reference<sup>1</sup></p>
```

The Pre Tag (Preformatted Text)

The <pre> tag displays text exactly as it is written in the code, including spaces and line breaks. The browser uses a monospace font. It is perfect for showing code snippets or ASCII art.

Example 2.10 – Preformatted Text

```
<pre>

function hello() {

console.log("Hello World");

}

</pre>
```

Chapter 2 – Practice Set

Practice 1: Create a page titled “My Resume”. Use h1 for your name, h2 for sections (Education, Skills, Experience), and paragraphs for details.

Practice 2: Add at least three external links (to your favorite websites) and one email link.

Practice 3: Insert an image of yourself or any placeholder image. Give it a proper alt attribute and set a reasonable width.

Practice 4: Write a short poem or address using the <br> tag for line breaks.

Practice 5: Create a page that demonstrates all six heading levels, bold, italic, underline, subscript and superscript.

Practice 6: Use the <pre> tag to display a small piece of Python or JavaScript code with proper indentation.

Important Questions – Chapter 2

Q1. What is the difference between an HTML element and an HTML attribute?

Q2. How many heading levels does HTML provide? Which one should be used for the main title of a page?

Q3. Why is the alt attribute important for images?

Q4. What is the difference between and <b>?

Q5. When should you use <br> instead of a new <p> tag?

Q6. Give two practical examples where <sub> or <sup> tags are useful.

Q7. What does the <pre> tag preserve that normal paragraphs do not?

Q8. Write the complete syntax of an image tag with src, alt, width and height attributes.

Chapter 3 – Creating A Page Layout

So far we have used tags that create visible content. Now we will learn tags that help us organize and structure that content into logical sections. This is the foundation of modern page layout.

Link Attributes

The <a> tag has several useful attributes beyond href.

Example 3.1 – Useful Link Attributes

```
<!-- Open link in a new tab -->
<a href="https://www.wikipedia.org" target="_blank">Wikipedia (new tab)</a>

<!-- Add a tooltip -->
<a href="about.html" title="Learn more about us">About</a>

<!-- Download a file instead of opening it -->
<a href="resume.pdf" download>Download My Resume</a>

<!-- Combine attributes -->
<a href="https://github.com" target="_blank" title="Visit my GitHub">
My GitHub Profile
</a>
```

target="_blank" opens the link in a new browser tab. This is very common for external links. Always consider adding rel="noopener" for security when using target="_blank".

The div Tag

The tag is a container (division). It is a block-level element that does not add any visual style by itself. Its main purpose is to group other elements together so you can style them or manipulate them with CSS and JavaScript later.

Think of as an invisible box. You put content inside the box, then later you can move, color, or resize the whole box at once.

Example 3.2 – Using div for Structure

```
<!DOCTYPE html>
<html>
<head>
<title>Page Layout with Div</title>
</head>
<body>

<div id="header">
<h1>My Website</h1>
<p>Welcome to my personal site</p>
```

```
</div>

<div id="navigation">
<a href="index.html">Home</a> |
<a href="about.html">About</a> |
<a href="contact.html">Contact</a>
</div>

<div id="main-content">
<h2>About Me</h2>
<p>I am learning HTML and loving it!</p>
</div>

<div id="footer">
<p>Copyright 2025 – All rights reserved</p>
</div>

</body>
</html>
```

TIP: In modern HTML we prefer semantic tags like <header>, <nav>, <main>, <section>, <article>, and <footer> instead of generic divs whenever possible. But understanding div is still essential because it is the most flexible container.

The span Tag

The tag is an inline container. Unlike (which is block-level and starts on a new line), sits inside a line of text without breaking the flow. It is perfect when you want to style or target just a few words inside a paragraph.

Example 3.3 – Using span

```
<p>This is a normal paragraph with
<span style="color: red;">this text highlighted in red</span>
and then normal text continues.</p>

<p>Price: <span class="price">■499</span> only!</p>

<p>My name is <span id="username">Aarav</span>.</p>
```

Quick comparison:

- **div** = block-level container (starts on a new line, takes full width by default)
- **span** = inline container (stays in the flow of text, only as wide as its content)

Both tags become truly powerful when you combine them with CSS (classes and IDs). For now, focus on understanding when to use each one.

Chapter 3 – Practice Set

Practice 1: Recreate the page layout example above (header, navigation, main content, footer) using div tags. Give each div a meaningful id.

Practice 2: Create a navigation bar with 4 links. Make external links open in a new tab using target="_blank".

Practice 3: Write a paragraph and use to highlight at least three different words or phrases (you can later style them with CSS).

Practice 4: Create a simple product card using a div. Inside it put a heading (product name), an image, a short description, and a price wrapped in a span.

Practice 5: Build a complete mini-website with three pages (Home, About, Contact) that link to each other using relative paths.

Important Questions – Chapter 3

Q1. What is the main purpose of the tag?

Q2. What is the difference between a block-level element and an inline element? Give one example of each.

Q3. When would you choose instead of ?

Q4. What does target="_blank" do in an anchor tag?

Q5. Why is it better to use semantic tags (<header>, <nav>, <footer>) instead of only divs when possible?

Q6. How can you make a link download a file instead of navigating to it?

Q7. Explain the difference between id and class attributes (even though we mainly use them later with CSS).

Chapter 4 – More Useful Tags (Bonus)

These tags appear in almost every real website. Learning them now will make you much more productive.

Lists – Unordered, Ordered and Description

Lists are one of the most common structures on the web.

Example 4.1 – Unordered List (bullets)

```
<ul>
<li>HTML</li>
<li>CSS</li>
<li>JavaScript</li>
<li>React</li>
</ul>
```

Example 4.2 – Ordered List (numbers)

```
<ol>
<li>Open VS Code</li>
<li>Create a new file</li>
<li>Write HTML code</li>
<li>Save and open in browser</li>
</ol>
```

Example 4.3 – Nested Lists

```
<ul>
<li>Frontend
<ul>
<li>HTML</li>
<li>CSS</li>
<li>JavaScript</li>
</ul>
</li>
<li>Backend
<ul>
<li>Node.js</li>
<li>Python</li>
</ul>
</li>
</ul>
```

Tables

Tables organize data into rows and columns. Use them for tabular data, not for page layout.

Example 4.4 – Simple Table

```
<table border="1">
<tr>
<th>Name</th>
<th>Age</th>
<th>City</th>
</tr>
<tr>
<td>Aarav</td>
<td>21</td>
<td>Delhi</td>
</tr>
<tr>
<td>Priya</td>
<td>23</td>
<td>Mumbai</td>
</tr>
</table>
```

tr = table row, **th** = table header cell, **td** = table data cell. The border attribute is for quick testing; real styling is done with CSS.

Forms Basics

Forms collect user input. Even a simple contact form needs these tags.

Example 4.5 – Basic Contact Form

```
<form action="/submit" method="post">
<label for="name">Name:</label><br>
<input type="text" id="name" name="name" required><br><br>

<label for="email">Email:</label><br>
<input type="email" id="email" name="email" required><br><br>

<label for="message">Message:</label><br>
<textarea id="message" name="message" rows="4" cols="40"></textarea><br><br>

<input type="submit" value="Send Message">
</form>
```

TIP: Common input types: text, email, password, number, checkbox, radio, file, date, submit, reset. Always pair inputs with proper labels for accessibility.

Chapter 4 – Practice Set

Practice 1: Create a page that shows your skills as an unordered list and your education timeline as an ordered list.

Practice 2: Build a simple timetable or comparison table with at least 3 columns and 4 rows.

Practice 3: Create a contact form with name, email, phone and message fields. Make name and email required.

Practice 4: Combine everything you have learned: create a complete personal portfolio page with header, navigation, about section, skills list, a small projects table, and a contact form in the footer.

Important Questions – Chapter 4

Q1. What is the difference between <ul> and ?

Q2. Which tags are used to create a table row, a header cell, and a data cell?

Q3. Why should tables not be used for page layout?

Q4. What is the purpose of the action and method attributes in a form?

Q5. Why is the label tag important for form inputs?

Quick Reference Cheat Sheet

Essential Document Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Page Title</title>
</head>
<body>
<!-- All visible content goes here -->
</body>
</html>
```

Most Used Tags at a Glance

Tag	Purpose	Example
<h1>–<h6>	Headings	Title
	Paragraph	Text
<a>	Link	Link
<img>	Image	<img src="x.jpg" alt="desc">
 	Line break	Line1 Line2
<hr>	Horizontal line	<hr>
/ <b>	Bold / Important	Bold
/ <i>	Italic / Emphasis	Italic
	Block container	...
	Inline container	...
/	Lists	Item
<table>	Table	<table><tr><td>...</td></tr></table>
<form>	Form	<form>...</form>
<input>	Form input	<input type="text">

What to Learn Next

You now have a solid foundation in HTML. The natural next steps are:

1. **CSS** – Learn how to style your pages (colors, fonts, flexbox, grid, responsive design).
2. **Semantic HTML** – Dive deeper into <header>, <nav>, <main>, <article>, <section>, <aside>, <footer>.
3. **Forms in depth** – Validation, different input types, accessibility.
4. **JavaScript basics** – Make your pages interactive.

Keep practicing every day. Build small projects. The more you write HTML, the more natural it becomes. Happy coding!

— End of Tutorial —